

# Improved Cryptanalysis of Forró via Trail Enumeration Puncturing

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# Background

- ChaCha, a variant of Salsa20, is a stream cipher designed by D. J. Bernstein in 2008 for high performance in software implementation.
- CHACHA: Secure Shell Protocol (SSH), Transport Layer Security (TLS 1.2, TLS 1.3), Secure/Multipurpose Internet Mail Extensions (S/MIME 4.0), Google Chrome, Android systems and Apple HomeKit.
- Forró [Cou+23], a successor to ChaCha, proposed at ASIACRYPT 2022 by Coutinho *et al.*, is designed to achieve a security level comparable to ChaCha.



# Design of Forró

- Forró is a **ARX**-based stream ciphers.
- ARX stands for:
  - **A** : Addition modulo  $2^{32}$  ( $\boxplus$ )
  - **R** : Bit Rotation ( $\lll r$ )
  - **X** : Exclusive-OR ( $\oplus$ )
- These simple operations provide both efficiency and security.

$$(a \boxplus b) \bmod 2^{32}, \quad a \lll r, \quad a \oplus b$$



- State of Forró :

$$V^0 = \begin{pmatrix} k_0 & k_1 & k_2 & k_3 \\ t_0 & t_1 & c_0 & c_1 \\ k_8 & k_9 & k_{10} & k_{11} \\ t_2 & t_3 & c_2 & c_3 \end{pmatrix}, \quad V^r = \begin{pmatrix} v_0^r & v_1^r & v_2^r & v_3^r \\ v_4^r & v_5^r & v_6^r & v_7^r \\ v_8^r & v_9^r & v_{10}^r & v_{11}^r \\ v_{12}^r & v_{13}^r & v_{14}^r & v_{15}^r \end{pmatrix}$$

Total 16 words. Each word 32 bits.

- $k_i$  : Key Words,  $c_i$  : Constants,  $t_i$  : Initial Values



## Sub-Round ( $\mathcal{SR}$ ) Function of Forró

- Each round of Forró consists of four Sub-Round ( $\mathcal{SR}$ ) functions.
- Each Sub-Round ( $\mathcal{SR}$ ) is applied on five words.
- $(p, q, r, s, t) \xrightarrow{\mathcal{SR}} (p'', q'', r'', s'', t'')$ .

$$\begin{aligned}
 s' &= s \boxplus t, & r' &= r \oplus s', & q' &= (q \boxplus r') \lll 10, \\
 p' &= p \boxplus q', & t'' &= t \oplus p', & s'' &= (s' \boxplus t'') \lll 27, \\
 r'' &= r' \boxplus s'', & q'' &= q' \oplus r'', & p'' &= (p' \boxplus q'') \lll 8.
 \end{aligned}$$

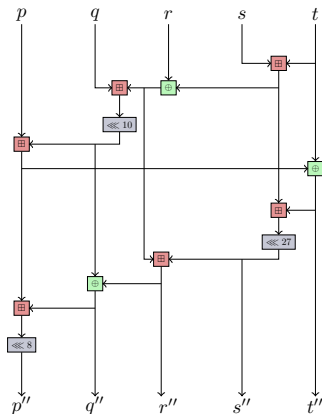


Figure: Diagram representing the Sub-Round ( $\mathcal{SR}$ ) Function



# PNB-Based Attack Framework

- At FSE 2008 [Aum+08], Aumasson et al. first introduced the PNB technique and applied it to ChaCha.
- Identify a Differential-Linear Distinguisher.
- Selecting a set of Probabilistic Neutral Bits (PNBs).
- Guessing the significant (non-PNB) key bits.
- Perform an exhaustive search over the PNB bits.



# Trail Enumeration Puncturing

- At EUROCRYPT 2025 [FT25], Flórez-Gutiérrez and Todo introduced a new key-recovery technique on ChaCha based on *trail enumeration puncturing*, later extended to Salsa at ASIACRYPT 2025 [FT26].
- This approach replaces the traditional PNB-based attack framework.
- The target bit is approximated by a pseudoboolean function.
- The correlation of the approximating function is then computed analytically via trail enumeration puncturing.



# Our Contributions

- ① We provide the first application of trail enumeration puncturing to Forró.
- ② We develop the first 3-round differential-linear distinguisher for Forró.
- ③ Combining this distinguisher with the partitioning technique and a distillation table, we reduce the time complexity of the best-known attack on 5-round Forró by about  $2^{11}$ .





Table: Comparison of attacks against 5-round Forró

| Target      | Data        | Time         | Notes                                              | Ref.      |
|-------------|-------------|--------------|----------------------------------------------------|-----------|
| Forró 5/256 | $2^{57}$    | $2^{158}$    | PNBs w/ Distinguisher                              | [Cou+23]  |
|             | $2^{52.36}$ | $2^{144.14}$ | PNBs w/ Multi-Step                                 | [SGD24]   |
|             | $2^{101.6}$ | $2^{136.94}$ | Puncturing w/ Improved Distinguisher               | This Work |
|             | $2^{101.6}$ | $2^{133.18}$ | Puncturing, Improved Distinguisher w/ Distillation | This Work |



# References

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*Thank You !*

